

PROPOSED MODEL & SYSTEM DESIGN

MCA 306 — Minor Project-II

Project Title: Placement OS — AI-Powered Multi-Tenant Campus Placement Management System

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1. SYSTEM OVERVIEW & ARCHITECTURAL SCOPE

Placement OS is an enterprise-grade, multi-tenant B2B Software-as-a-Service (SaaS) platform engineered to unify, automate, and algorithmically optimize the end-to-end campus placement lifecycle across higher education institutions. The system integrates four specialized user roles — **Students**, **Corporate Recruiters (Companies)**, **College Placement Authorities (TPOs)**, and **Platform Super-Administrators** — operating under strict row-level tenant isolation, rate-limited API access, and granular Role-Based Access Control (RBAC).

The core computational pillars of Placement OS encompass:

- Dual-Engine AI ATS Resume Screening:** Integrating mathematical n-gram TF-IDF vector space modeling with NLP skill-dictionary parsing and Generative AI qualitative feedback.
- Cryptographic Blockchain-Inspired Offer Verification:** SHA-256 sequential hash-chaining binding student, company, and remuneration credentials with public QR validation.
- Supervised Machine Learning Placement Predictor:** Logistic Regression classification model calculating probability-of-placement based on academic and skill features.
- Multi-Channel Asynchronous Event Pipeline:** Celery-Redis task workers dispatching transactional emails, in-app alerts, and automated WhatsApp notifications.

2. SYSTEM ARCHITECTURE & BLOCK DIAGRAM

1. Client & Presentation Layer

Tech: Next.js 16.3, React 19, TypeScript, Tailwind CSS, Lucide Icons, Sonner

Modules: Responsive Student Portal, Recruiter Job Workspace, College TPO Roster & CSV Ingestion, SuperAdmin Verification Console.

2. API Gateway & Middleware Layer

Tech: Django REST Framework, SimpleJWT (Token Rotation + Blacklist), CORS Middleware

Security: TenantMiddleware (Request Scoping), TenantRateThrottle (User-ID Keyed Quotas), Brute-Force Lockout Defense.

3. Business & Application Services Layer

Tech: Python 3.12, Django ORM, Celery 5.4 Async Task Queues, Celery Beat Schedulers

Apps: accounts, students, companies, jobs, interviews, ai_module, data_science, blockchain_module, analytics, learn, notifications, billing.

4. AI / ML & Cryptographic Engine

Tech: Scikit-Learn (TF-IDF), Gemini 1.5 Flash API, pdfplumber/PyPDF2, hashlib (SHA-256), qrcode

Services: 3-tier ATS scoring, Gemini feedback generation, LogReg placement probability, Offer hash-chaining & QR encoding.

5. Data Persistence, Caching & File Storage Layer

Relational Store: PostgreSQL (Production with Connection Pooling) / SQLite3 (Development)

In-Memory Cache & Broker: Redis 5.0 (Session Cache, Throttle Storage, Celery Message Broker)

Media Storage & Static Delivery: Cloudinary Storage (CDN Assets, PDF Resumes, Offer QR Codes) / WhiteNoise

Figure 2.1: Multi-Tiered Modular System Architecture Block Diagram of Placement OS

3. COMPREHENSIVE TECHNOLOGY STACK SPECIFICATIONS

Architectural Layer	Technology / Framework	Functional Responsibility & Rationale
Frontend Framework	Next.js 16.3 (React 19, TypeScript)	Server-Side Rendering (SSR), App Router architecture, type-safe API communication, reactive dashboard state management.
Styling & Components	Tailwind CSS 4.3.3, Lucide-React	Utility-first responsive styling, dark/light accessibility, component composition for enterprise UX.
Backend API Core	Django 5.x, Django REST Framework	High-throughput RESTful API, robust ORM, automatic schema serialization, enterprise middleware pipeline.
Authentication & 2FA	SimpleJWT, Email OTP Service	Stateless JWT token rotation (30-min access, 1-day refresh), mandatory email-based 2-Factor Authentication (MFA).
Database & ORM	PostgreSQL, dj-database-url	ACID-compliant relational persistence, composite indexing, JSONField attributes for dynamic skill mapping.
Task Queue & Caching	Celery 5.4, Redis 5.0, Celery Beat	Asynchronous background PDF parsing, automated transactional email dispatch, daily analytics digests.
Natural Language Processing	Scikit-Learn (TF-IDF Vectorizer)	N-gram (1,2) document vectorization and Cosine distance similarity evaluation between resumes and job specs.
Generative AI Engine	Google Gemini 1.5 Flash REST API	Automated prompt-engineered synthesis of qualitative, constructive resume improvement feedback.
Machine Learning Core	Joblib, Scikit-Learn (Logistic Regression)	Feature-engineered candidate probability classification modeling with mathematical fallback heuristic.
Document & PDF Parsing	pdfplumber, PyPDF2	Extraction of raw text from multi-page PDF resumes, magic byte inspection (b"%PDF-"), ownership validation.
Blockchain & Hashing	Hashlib (SHA-256), qrcode 7.4.2	Cryptographic hash-chain linking of consecutive offer letters, generation of verification QR matrices.
Multi-Channel Alerts	Twilio API, Brevo SMTP / API	Direct student WhatsApp status updates, transactional OTP distribution with 2-minute expiration.
API Schema & Docs	drf-spectacular (OpenAPI 3.0)	Dynamic OpenAPI schema generation powering interactive Swagger-UI and ReDoc API explorers.
Observability & Logs	Python-JSON-Logger, Sentry SDK	Centralized structured JSON logging across all backend modules, PII-scrubbed distributed error tracing.

4. ENTITY-RELATIONSHIP (ER) SCHEMA DESIGN

Placement OS is governed by a normalized relational database schema comprising 18 interconnected models across functional apps:

Figure 4.1: Relational Entity Structure and Cardinality Mapping of Placement OS

5. DATA FLOW DIAGRAMS (DFDS)

5.1 Context Diagram (Level-0 DFD)

The Context DFD establishes the environmental boundary of Placement OS, mapping information exchanges between external entities and the centralized system:

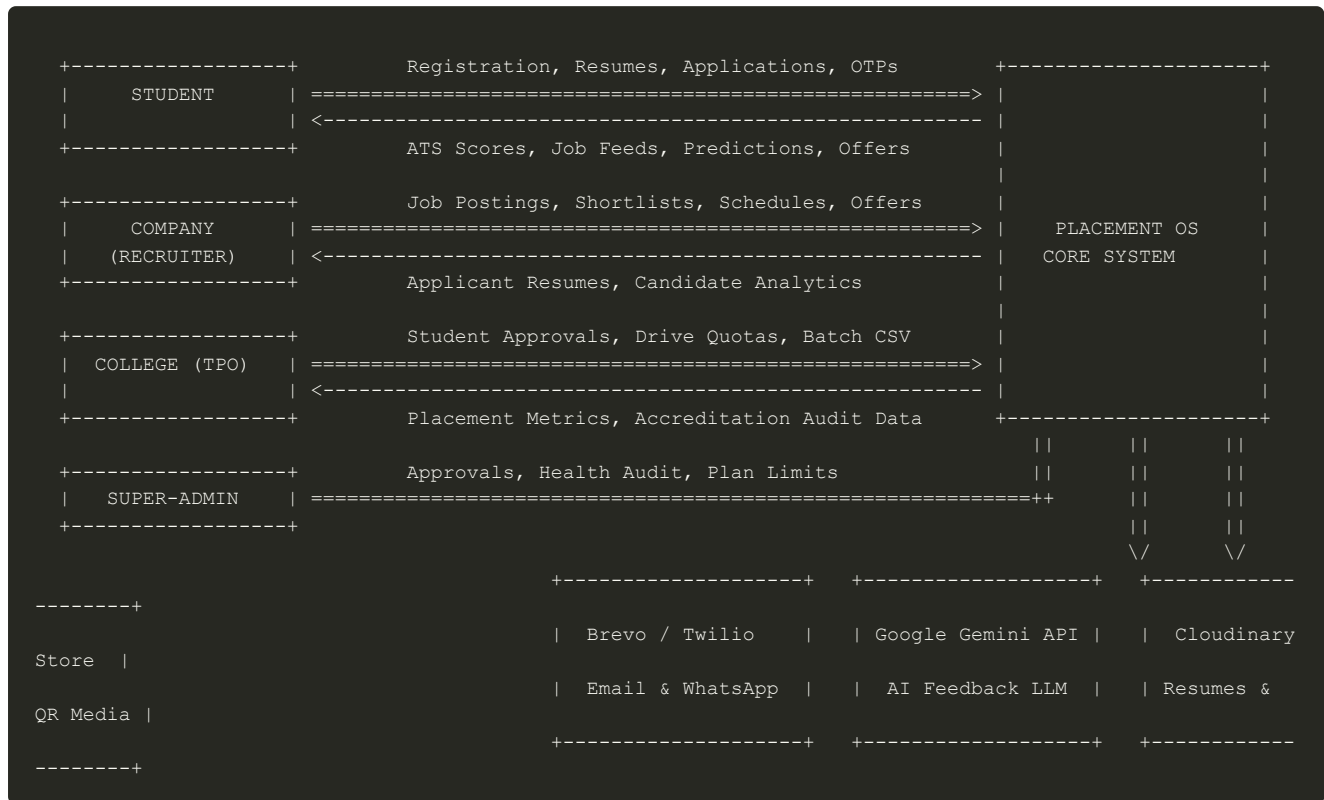


Figure 5.1: Level-0 Context Data Flow Diagram of Placement OS

5.2 Level-1 DFD — Functional Decomposition

The Level-1 DFD decomposes the system into 7 primary computational processes:

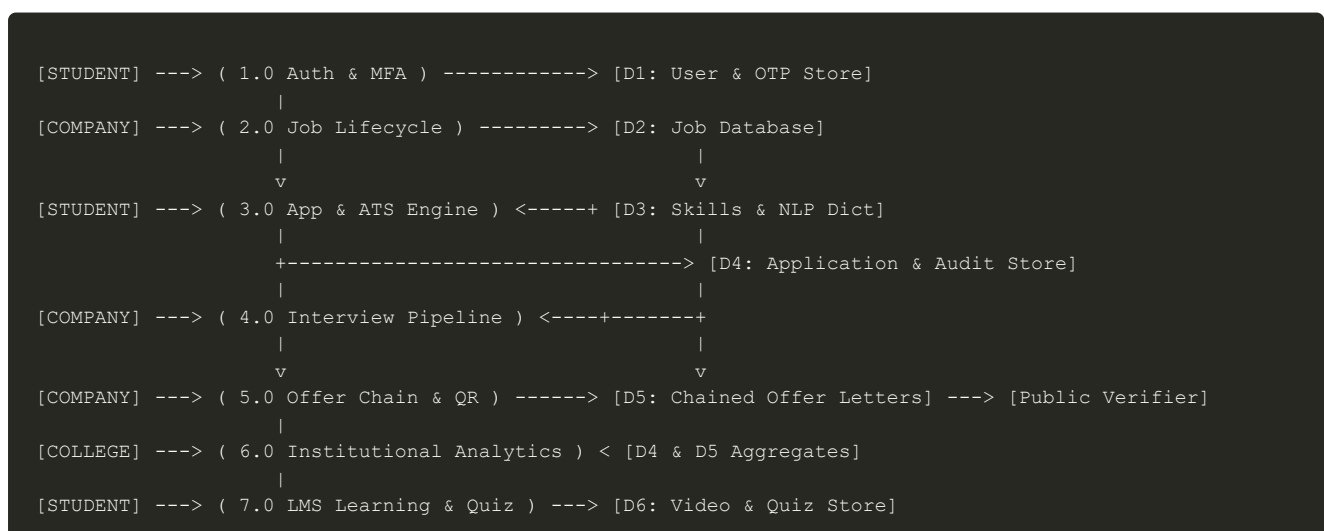


Figure 5.2: Level-1 Functional Decomposition Data Flow Diagram

5.3 Level-2 DFD — Subsystem 3.0: Application Processing & ATS Scoring

```
[Candidate Application + PDF Resume]
|
v
( 3.1 Eligibility Check ) -----> [Branch ∈ eligibility_branch && CGPA >= min_cgpa]
| (Pass)
v
( 3.2 Magic Byte & Text Extraction ) ---> [pdfplumber / PyPDF2 Stream Reader]
| (Extracted Text)
+-----+
|                                     |
v                                     v
( 3.3 NLP Skill Extraction )         ( 3.4 TF-IDF Vectorization )
Match against [Skills Dictionary]    n-gram (1,2) Cosine Distance vs JD
|                                     |
+-----+
|
v
( 3.5 Weighted ATS Scoring )
50% Cosine + 35% Skill Match + 15% Qual
|
v
( 3.6 Gemini AI Feedback )
Generates 2-3 sentence remediation
|
v
[Save Application + ATS Score + Audit Log]
```

Figure 5.3: Level-2 Detailed DFD for ATS Scoring and Feedback Subsystem

6. CORE PROCESS FLOWCHARTS

6.1 User Registration, Two-Factor Authentication & RBAC Onboarding

```
[Start] --> [User Submits Registration Form] --> [Validate Email, Pass, Role]
|
+--> Email Exists? --(Yes)--> [Display Conflict Error] --> [Retry]
|
+--(No)--> [Create User Record (is_verified=False)]
|
|   v
|   [Generate 6-Digit Cryptographic OTP]
|   |
|   |   v
|   |   [Dispatch Celery Task: send_otp_email_task (2-5 min expiry)]
|   |   |
|   |   |   v
|   |   |   [User Inputs OTP on Client Screen]
|   |   |   |
|   |   |   |   +--> OTP Valid & Attempts < 3? --(No)--> [Increment Attempts / Lockout]
|   |   |   |   |
|   |   |   |   +--(Yes)--> [Set is_verified = True, Mark OTP Used]
|   |   |   |   |
|   |   |   |   |   v
|   |   |   |   |   [Role-Based Redirection]
|   |   |   |   |   |-- STUDENT --> [Student Profile Setup (CGPA, Branch, Resume)]
|   |   |   |   |   |-- COMPANY --> [Company Workspace Setup (GST, HR Details)]
|   |   |   |   |   |-- COLLEGE --> [TPO Institutional Onboarding & Verification]
|   |   |   |   |   |-- ADMIN --> [SuperAdmin Console]
```

Figure 6.1: Registration, Multi-Factor Authentication (OTP) and Onboarding Flowchart

6.2 Cryptographic Offer Letter Generation & Public QR Verification

```
[Company Selects Candidate] --> [Input CTC, Designation, Joining Date]
|
|   v
|   [Retrieve Previous Offer's SHA-256 Hash]
|   |-- If Genesis (First Offer) --> previous_hash = "000...000" (64 zeros)
|   |-- Else --> previous_hash = last_offer.hash_sha256
|   |
|   |   v
|   |   [Construct Deterministic Hash String: Student | Company | Role | CTC | Date | Content | PrevHash]
|   |   |
|   |   |   v
|   |   |   [Compute SHA-256 Digest] --> [Save OfferLetter Record with Hash Pointer]
|   |   |   |
|   |   |   |   v
|   |   |   |   [Generate Public QR Code: https://placement-os/verify/offer/{offer_id}/]
|   |   |   |   |
|   |   |   |   |   v
|   |   |   |   |   [Update Application Status = OFFERED] --> [Student Receives Tamper-Proof Offer Letter]
|   |   |   |   |
|   |   |   |   |   --- PUBLIC VERIFICATION FLOW ---
|   |   |   |   |   [Verifier Scans QR / Opens URL] --> [Fetch Offer Record by UUID]
|   |   |   |   |   |
|   |   |   |   |   |   v
|   |   |   |   |   |   [Recompute SHA-256 Hash using Stored Field Payload]
|   |   |   |   |   |   |
|   |   |   |   |   |   |   +--> Stored Hash == Recomputed Hash?
|   |   |   |   |   |   |   |--(No) --> [DISPLAY: RED ALERT - Document Tampered / Invalid]
|   |   |   |   |   |   |   |--(Yes) --> [DISPLAY: GREEN BADGE - Authenticated Offer Verified]
```

v
[Increment verified_count Metric]

Figure 6.2: Blockchain-Inspired Sequential Hash-Chain Generation & QR Verification Flowchart

7. CORE MATHEMATICAL FORMULATIONS & ALGORITHMS

7.1 Hybrid ATS Resume Scoring Algorithm

$$\text{ATS Score} = \text{Big}(0.50 \times \text{cosine_sim} \times 100) + \text{Big}(0.35 \times \text{skills_ratio} \times 100) + \text{Big}(0.15 \times \text{qual_factor} \times 100)$$

Where: $\text{cosine_sim} = \frac{\mathbf{v}_{\text{resume}} \cdot \mathbf{v}_{\text{job}}}{\|\mathbf{v}_{\text{resume}}\| \|\mathbf{v}_{\text{job}}\|}$, $\text{skills_ratio} = \frac{|\mathbf{K}_{\text{resume}} \cap \mathbf{K}_{\text{job}}|}{|\mathbf{K}_{\text{job}}|}$, $\text{qual_factor} = \begin{cases} 1.0 & \text{if branch eligible} \\ 0.5 & \text{otherwise} \end{cases}$

```
ALGORITHM: CalculateHybridATSScore
INPUT: resume_pdf_stream, job_record, student_profile
OUTPUT: { ats_score: Float, matched_skills: List, missing_skills: List, ai_feedback: String }

BEGIN
    // Step 1: Text extraction and validation
    raw_text = ExtractTextWithFallback(resume_pdf_stream)
    IF raw_text IS EMPTY THEN RETURN { ats_score: 0.0, error: "Unreadable PDF" }

    // Step 2: N-gram TF-IDF Vector Space Modeling
    vectorizer = TfidfVectorizer(ngram_range=(1, 2), stop_words='english')
    tfidf_matrix = vectorizer.fit_transform([raw_text, job_record.description])
    cosine_sim = cosine_similarity(tfidf_matrix[0:1], tfidf_matrix[1:2])[0][0]

    // Step 3: NLP Dictionary Skill Extraction
    extracted_skills = MatchSkillDictionary(raw_text, SKILLS_DICTIONARY)
    required_skills = job_record.required_skills
    matched_skills = extracted_skills INTERSECT required_skills
    missing_skills = required_skills DIFFERENCE extracted_skills
    skill_ratio = LENGTH(matched_skills) / MAX(LENGTH(required_skills), 1)

    // Step 4: Qualification & Cutoff Computation
    qual_factor = (student_profile.branch IN job_record.eligibility_branch) ? 1.0 : 0.5

    // Step 5: Final Weighted Aggregation
    composite_ats = (0.50 * cosine_sim * 100) + (0.35 * skill_ratio * 100) + (0.15 * qual_factor * 100)
    final_score = CLAMP(ROUND(composite_ats, 2), 0.0, 100.0)

    // Step 6: Asynchronous Generative AI Feedback (Gemini API)
    feedback_text = GenerateGeminiFeedback(raw_text, matched_skills, missing_skills)

    RETURN { ats_score: final_score, matched_skills, missing_skills, ai_feedback: feedback_text }
END
```

7.2 Sequential Hash-Chain Generation & Integrity Verification Algorithm

```
ALGORITHM: ComputeSequentialOfferHash
INPUT: candidate_name, company_name, designation, salary_ctc, joining_date, offer_text
OUTPUT: { sha256_digest: String, previous_hash: String }

BEGIN
    last_offer = OfferLetter.objects.order_by('-issued_at').first()
    previous_hash = (last_offer != NULL) ? last_offer.hash_sha256 : REPEAT("0", 64)

    payload_string = CONCAT_WITH_PIPE(
        candidate_name, company_name, designation, salary_ctc, joining_date, offer_text, previous_hash
    )
```

```
    sha256_digest = SHA256(UTF8_ENCODE(payload_string)).HEX_DIGEST()  
    RETURN { sha256_digest, previous_hash }  
END
```

8. DETAILED MODULE-WISE SYSTEM ARCHITECTURE

Django Module	Database Entities & Schema Keys	Functional Logic & Business Rules
accounts	User, CollegeProfile, CompanyProfile, OTPVerification, LoginHistory	Manages 4-tier RBAC, JWT issuance, 2-minute MFA verification, 15-minute brute-force lockout after 5 failed logins, GDPR account deletion.
students	StudentProfile, Certification, Project	Manages academic credentials, streak gamification, PDF resume validation. Auto-invalidates is_verified status if CGPA or branch is altered.
jobs	Job, Application, ApplicationAuditLog	Job posting CRUD, candidate application filtering, mandatory rejection reason capture (Feature 52), ZIP bulk resume download.
interviews	Interview, InterviewPerformance	Multi-round scheduling (Online/Offline), panel details management, structured scoring, and outcome classification (Selected/Rejected/On-Hold).
ai_module	*NLP Engine (Stateless)*	Dual-engine ATS scoring, pdfplumber text parser, skills_dictionary matcher, Gemini 1.5 Flash API feedback synthesizer.
blockchain_module	OfferLetter	Cryptographic SHA-256 sequential hash pointer chaining, automated QR image generation, public zero-auth verification endpoint.
data_science	PlacementPrediction	Logistic Regression model (logreg_v1.pkl) calculating student selection probability across open jobs with weighted rule-based fallback.
analytics	SupportTicket, PlacedStudent, Notification	Aggregates branch-wise placement percentages, median CTC metrics, daily digest Celery tasks, support ticket tracking.
learn	LearnModule, TestResult	Embedded YouTube LMS lectures, randomized 15-MCQ test engine, automated scoring, Gemini AI dynamic question generator.
notifications	Notification (app-wide)	Typed user alerts (info, success, warning, error), real-time unread counts, integration with Twilio WhatsApp messaging.
billing	Plan, Subscription	Multi-tenant B2B SaaS subscription management, job posting quota limits, college student roster enforcement, Stripe webhook listener.

9. MULTI-LAYERED SECURITY ARCHITECTURE

- Layer 1 — Transport Security:** Strict HTTPS/TLS enforcement, HSTS (31,536,000s), Secure & HttpOnly Cookie flags, X-Frame-Options: DENY.
- Layer 2 — Authentication & Token Lifecycle:** SimpleJWT stateless tokens (30-min access, 1-day refresh) with automated rotation and token blacklisting.
- Layer 3 — Two-Factor MFA Hardening:** Mandatory email OTP for logins with 2-minute expiration and 3-attempt brute-force invalidation.
- Layer 4 — Role-Based Access Control:** Strict permission classes (IsAuthenticated, IsCompany, IsCollege, IsAdmin) on all DRF viewsets.
- Layer 5 — Logical Multi-Tenancy Isolation:** TenantMiddleware resolves tenant context per-request, enforcing row-level database QuerySet isolation.

6. **Layer 6 — Tenant-Scoped Rate Throttling:** Custom TenantRateThrottle keys buckets by Tenant ID, insulating tenants from noisy-neighbor quota exhaustion.
7. **Layer 7 — Data Integrity & Auditability:** SHA-256 chained hashes for digital offers, immutable ApplicationAuditLog tracking all status transitions.
8. **Layer 8 — Centralized Observability:** Structured JSON logging via python-json-logger, PII-sanitized error monitoring via Sentry.

Figure 9.1: 8-Layer Defense-in-Depth Security Matrix of Placement OS

10. FRONTEND ARCHITECTURE & COMPLETE ROUTE MAP

The Next.js 16 client application exposes 28 distinct, role-guarded routes structured within the App Router paradigm:

Route Path	Target User Role	Functional Responsibility & Component Purpose
/	Public	Platform landing page displaying feature highlights, metrics, and testimonials.
/login	Public	Two-step authentication interface supporting email, password, and MFA OTP inputs.
/signup	Public	Multi-role registration interface with Clause 7 & 8 legal consent captures.
/forgot-password	Public	Self-service password recovery with email OTP verification.
/dashboard	Authenticated	Intelligent role-based redirect router dispatching users to designated portals.
/student-dashboard	Student	Candidate portal with ATS score display, job search, interview tracker, streak metrics.
/company-dashboard	Company	Recruiter workspace for job posting creation, applicant review, interview scheduling.
/college-dashboard	College / TPO	Institutional dashboard displaying student rosters, CSV batch ingestion, placement KPIs.
/admin	SuperAdmin	System-wide administrative health, tenant management, and platform metrics.
/admin/approvals	SuperAdmin	Verification and approval console for pending Company and College tenant registrations.
/tpo/approvals	College / TPO	Internal student profile verification portal for academic credential authorization.
/jobs	Authenticated	Searchable and filterable job board (branch, CTC, skills, location).
/applications	Student	Personal job application tracker with real-time status badges.
/applications/[id]	Student / Recruiter	Detailed application lifecycle view displaying ATS breakdown, audit timeline, reasons.
/interviews	Authenticated	Calendar view of upcoming online/offline technical and HR interview rounds.
/companies	Authenticated	Searchable directory of verified corporate recruiters.
/companies/[id]	Authenticated	Public corporate profile displaying active job drives and recruiter background.
/onboarding/student	Student	Profile completion wizard capturing CGPA, branch, semester, and resume PDF.
/onboarding/company	Company	Recruiter setup wizard capturing GSTIN, company bio, and verification docs.
/onboarding/college	College	TPO setup wizard capturing institutional accreditation, TPO contact, and branding.
/about, /contact	Public	Institutional information, help desk submission, and platform overview.
/privacy, /terms	Public	Legal compliance documentation, GDPR privacy policies, terms of service.

Route Path	Target User Role	Functional Responsibility & Component Purpose
/dmca, /tenant-agreement	Public	Intellectual property notices and B2B SaaS institutional terms of agreement.

11. RESTFUL API ENDPOINT DIRECTORY

API Endpoint Pattern	HTTP Method	Backend DRF View & Functional Purpose
/health/, /ready/	GET	HealthCheckView: System liveness and DB/Redis readiness check.
/verify/offer/<uuid:id>/	GET	VerifyOfferView: Public zero-auth QR tamper verification endpoint.
/api/v1/accounts/register/	POST	RegisterView: Registers new tenant/user and dispatches OTP email.
/api/v1/accounts/login/	POST	LoginView: Password verification and 2FA OTP challenge initiation.
/api/v1/accounts/login/verify-mfa/	POST	VerifyLoginMFView: Validates 2FA OTP and issues SimpleJWT tokens.
/api/v1/token/refresh/	POST	TokenRefreshView: Rotates refresh token and issues fresh access token.
/api/v1/students/me/	GET, PUT	StudentMeView: Retrieves and updates candidate profile credentials.
/api/v1/students/resume/upload/	POST	ResumeUploadView: Magic byte inspection and async resume text parsing.
/api/v1/jobs/	GET, POST	JobListCreateView: Lists active jobs (GET) or posts job with quota check (POST).
/api/v1/jobs/<uuid:id>/apply/	POST	ApplyJobView: Validates eligibility, creates app, and triggers ATS task.
/api/v1/jobs/<uuid:id>/applicants/	GET	ApplicantsPerJobView: Returns candidate list ranked by ATS score.
/api/v1/jobs/<uuid:id>/applicants/download/	GET	BulkResumeDownloadView: Generates and streams dynamic ZIP of all resumes.
/api/v1/interviews/	GET, POST	ScheduleInterviewView: Schedules round (POST) or lists interviews (GET).
/api/v1/blockchain/generate/	POST	GenerateOfferView: Chains SHA-256 hash and generates digital offer QR code.
/api/v1/ds/placement-probability/	GET	PlacementProbabilityView: Computes ML probability of candidate selection.
/api/v1/ds/recommended-jobs/	GET	RecommendedJobsView: Computes TF-IDF similarity recommendations for student.
/api/v1/learn/modules/	GET	LearnListView: Lists curated video learning modules and resources.
/api/v1/learn/test/submit/	POST	TestSubmitView: Evaluates MCQ responses and increments streaks/points.
/api/v1/analytics/overview/	GET	AnalyticsOverviewView: Delivers cached placement metrics and KPI counters.

API Endpoint Pattern	HTTP Method	Backend DRF View & Functional Purpose
/api/v1/billing/checkout/	POST	CheckoutView: Initiates Stripe checkout session for plan upgrade.